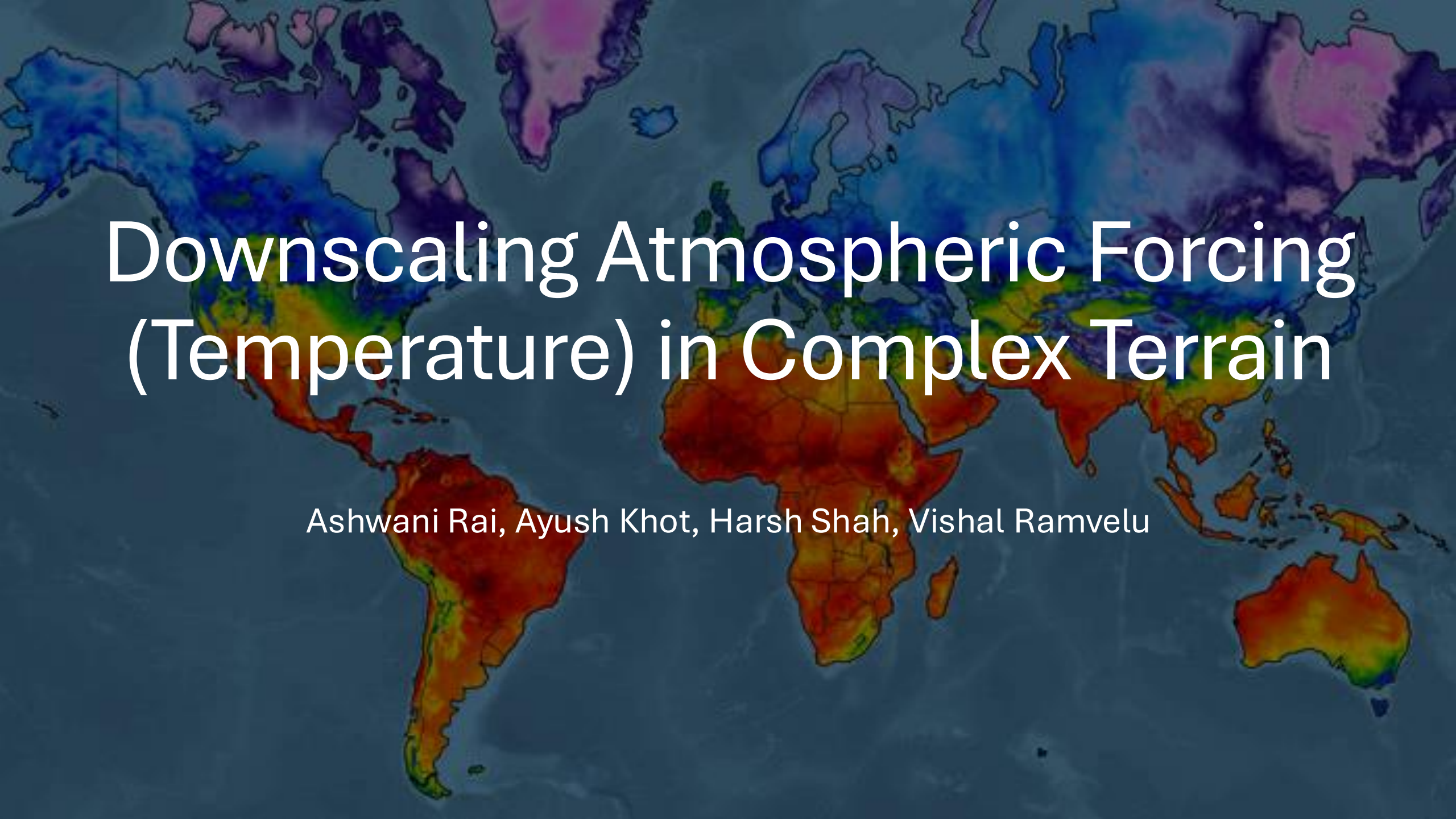
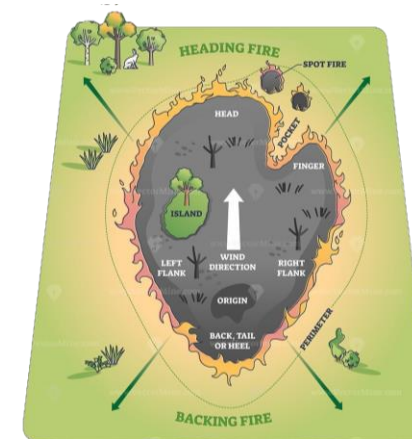
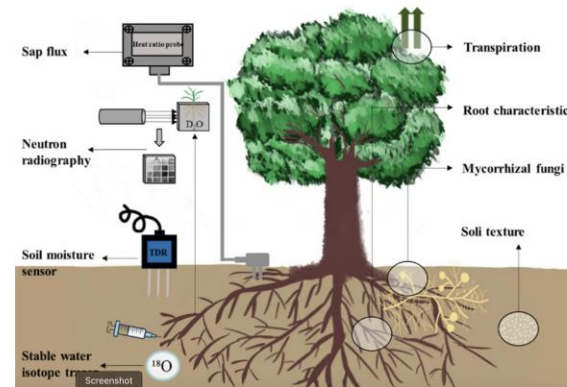
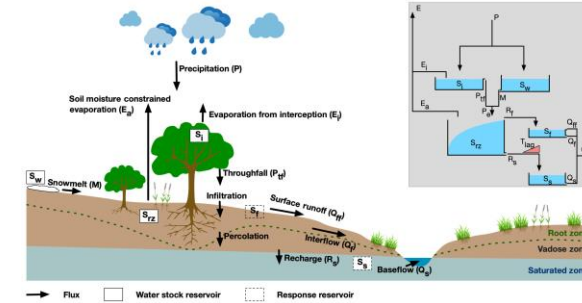
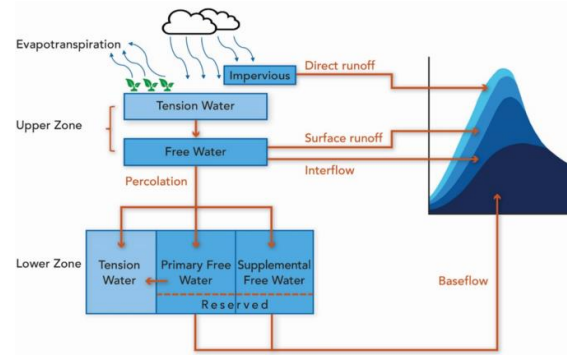
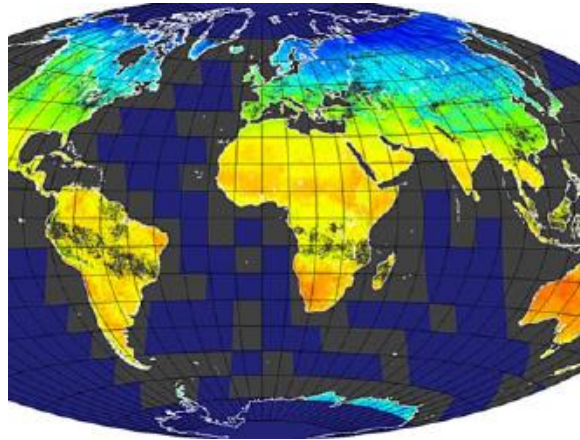


A world map with a color gradient representing atmospheric forcing. The colors range from dark blue/purple in the high northern latitudes to red/orange in the tropics and southern latitudes. The map is centered on the Atlantic Ocean, showing the Americas on the left and Europe/Africa on the right.

Downscaling Atmospheric Forcing (Temperature) in Complex Terrain

Ashwani Rai, Ayush Khot, Harsh Shah, Vishal Ramvelu

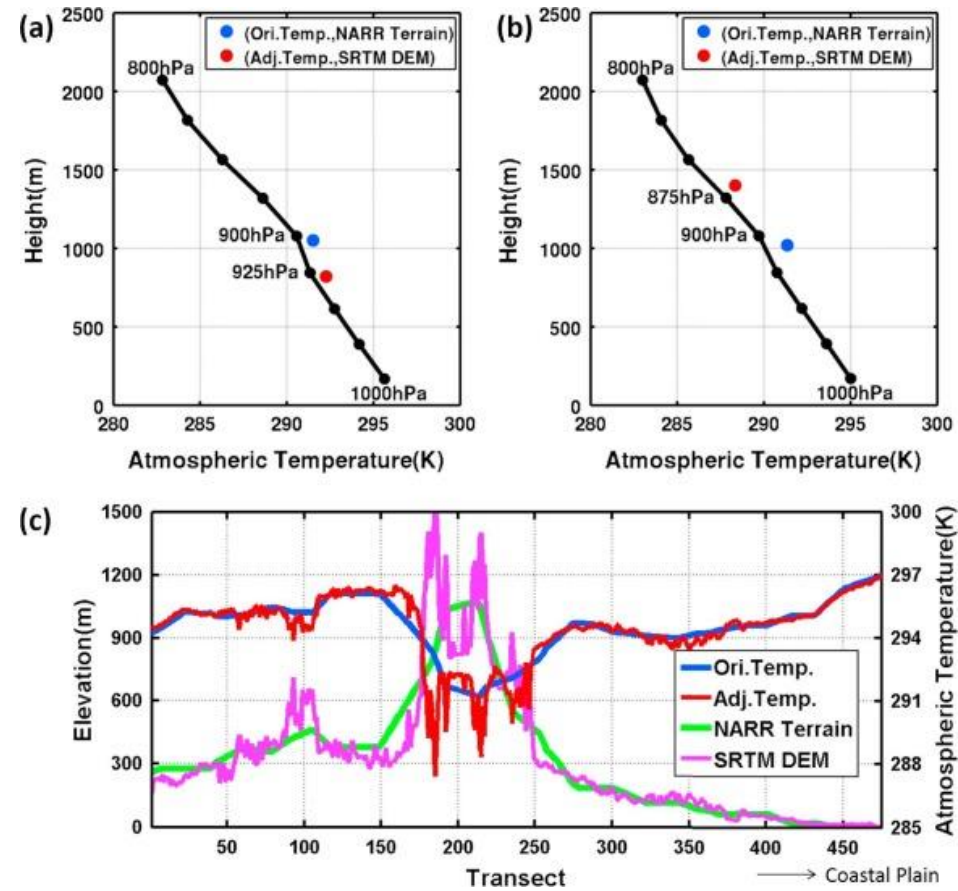
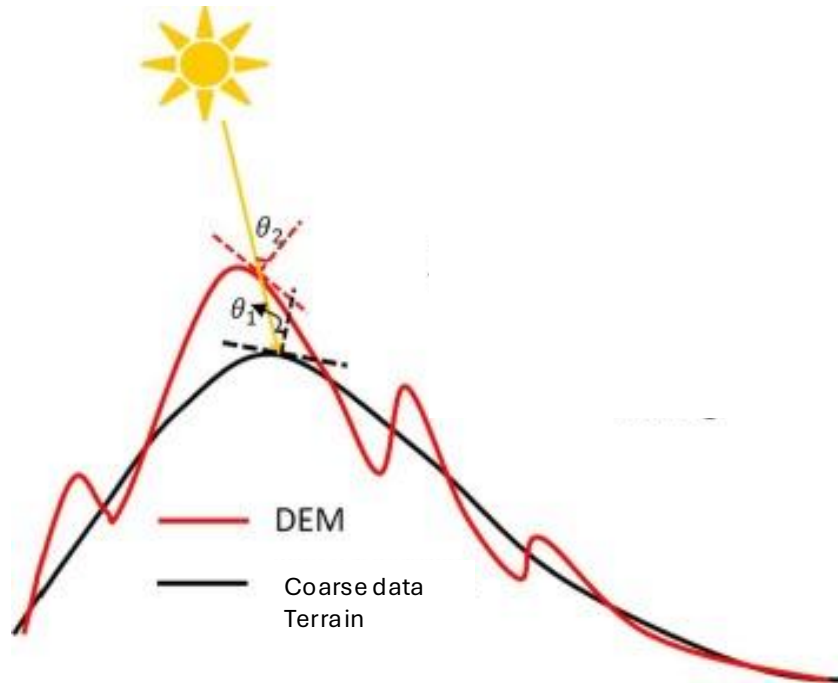
Need for Downscaling



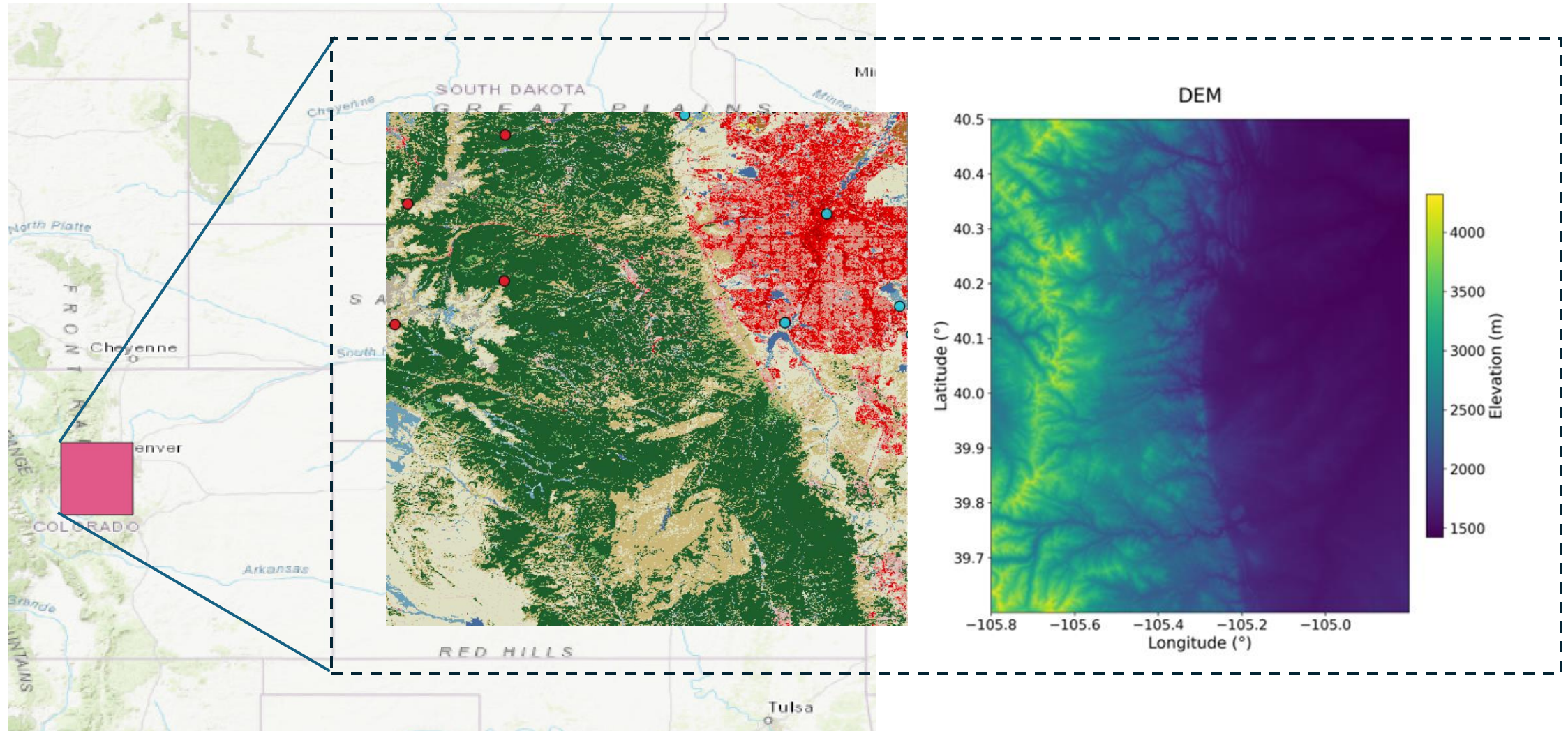
EO atmospheric forcing data:
Coarse spatial resolution
(~kms)

Variety of regional models need high resolution
and good quality forcing data as input (~100 m)

Terrain: Influencing the downscaling physics



Study region



Related Work

Traditional downscaling methods include interpolation, lapse-rate correction, regression, and bias-corrected spatial disaggregation. These approaches are efficient but often struggle to represent nonlinear temperature controls in complex terrain.

Deep Learning methods:

- **DeepSD** introduced CNN-based super-resolution for climate downscaling.
- **Afshari et al. (2023)** downscaled SEVIRI LST to WRF near-surface air temperature using deep learning.
- **Kheir et al. (2023)** proposed an improved deep learning procedure for statistical climate downscaling.
- **Tan et al. (2019)** used CNNs to retrieve LST from coarse AMSR2 observations.
- **Liu et al. (2023)** used a terrain-guided attention network to include DEM information in temperature downscaling.
- **Li et al. (2025)** used attention mechanisms and Google Earth Engine for ERA5 LST downscaling.
- **SRDA-Net (2025)** targets MODIS LST downscaling for high-resolution data generation.



Related Work

Traditional downscaling methods include interpolation, lapse-rate correction, regression, and bias-corrected spatial disaggregation. These approaches are efficient but often struggle to represent nonlinear temperature controls in complex terrain.

Deep Learning methods:

This project proposes a **dual-attention residual architecture** for terrain-aware MODIS LST downscaling using **Channel and Spatial Attention** to adaptively focus on the most important predictors and spatial regions.

- **Tan et al. (2019)** used CNNs to retrieve LST from coarse AMSR2 observations.
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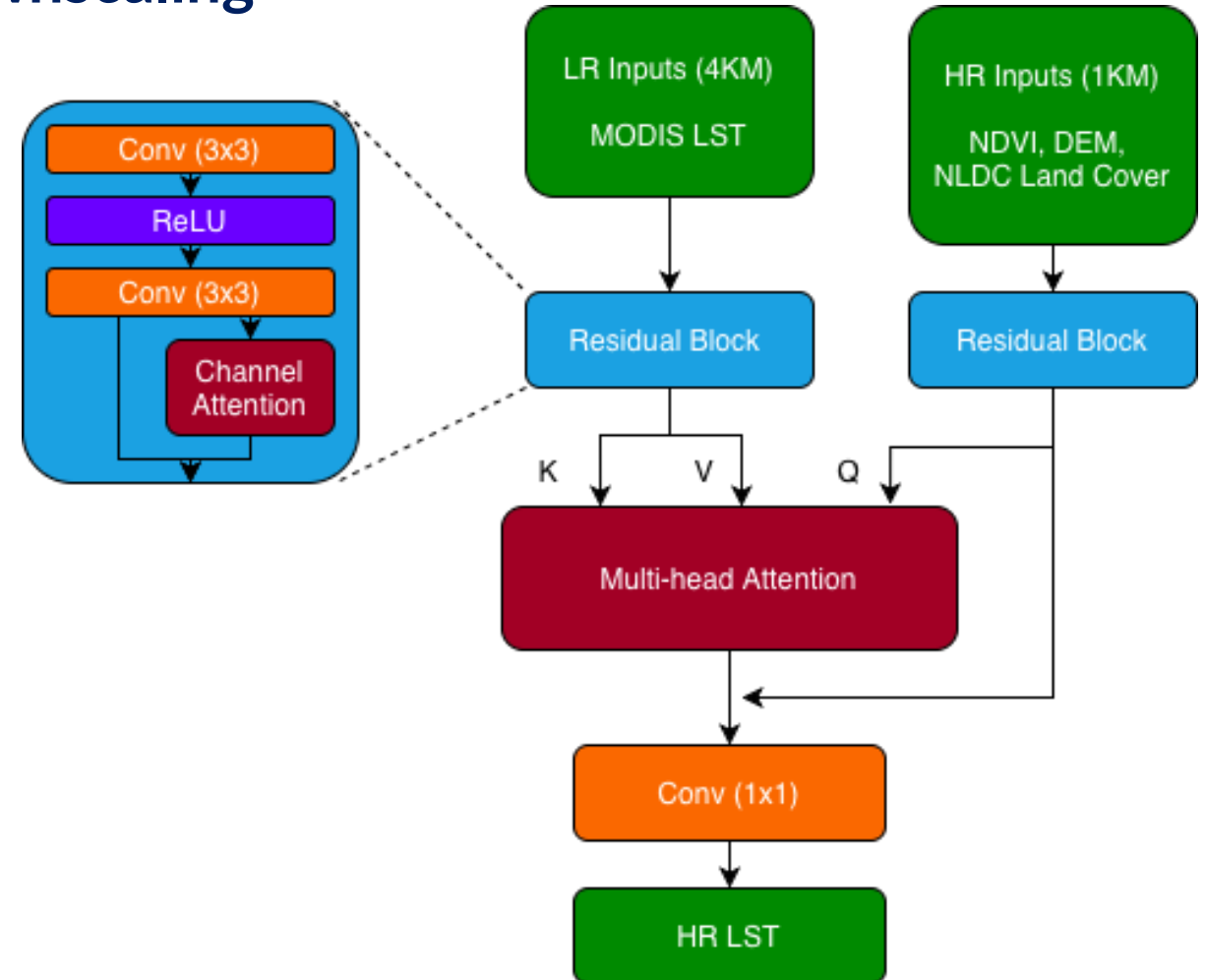
Attention augmented CNN based downscaling

- Inputs

- Low Resolution MODIS Temperature
- High Resolution DEM, NDVI, NLDC Land Cover

- Dual Attention

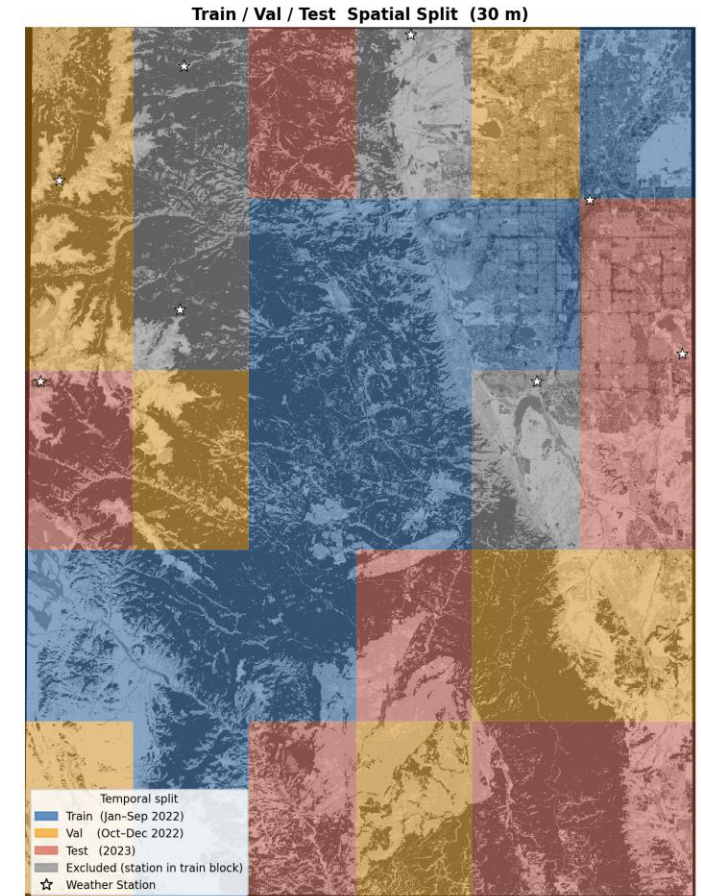
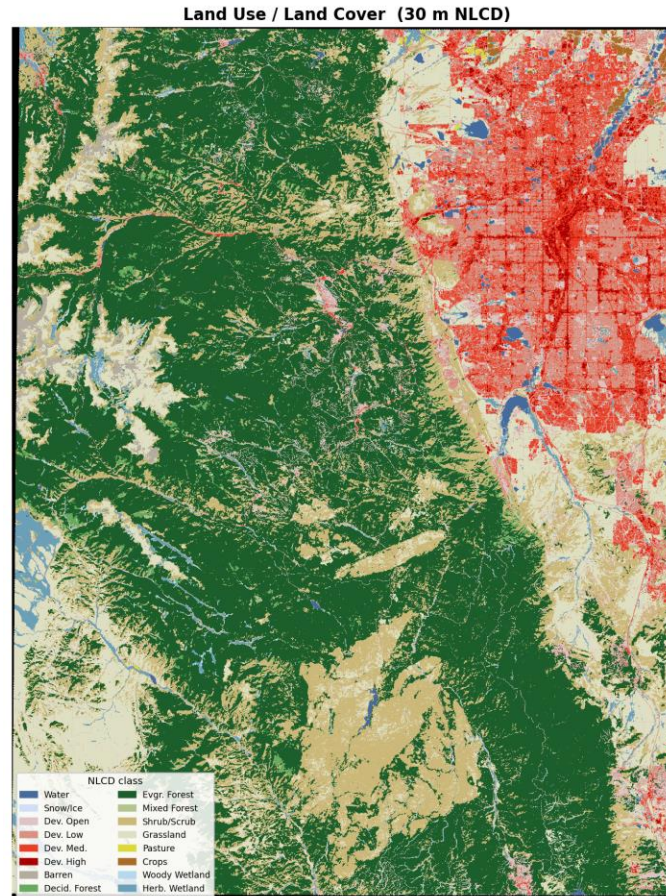
- Channel Attention:
 - Gating Attention
 - Feature importance
- Fusion Attention:
 - Junction between LR and HR
 - HR queries combined with LR Key-Value pairs



Training Details

Split data both temporally and spatially

- Train: Jan-Sep 2022
- Val: Oct-Dec 2022
- Test: Jan-Dec 2023



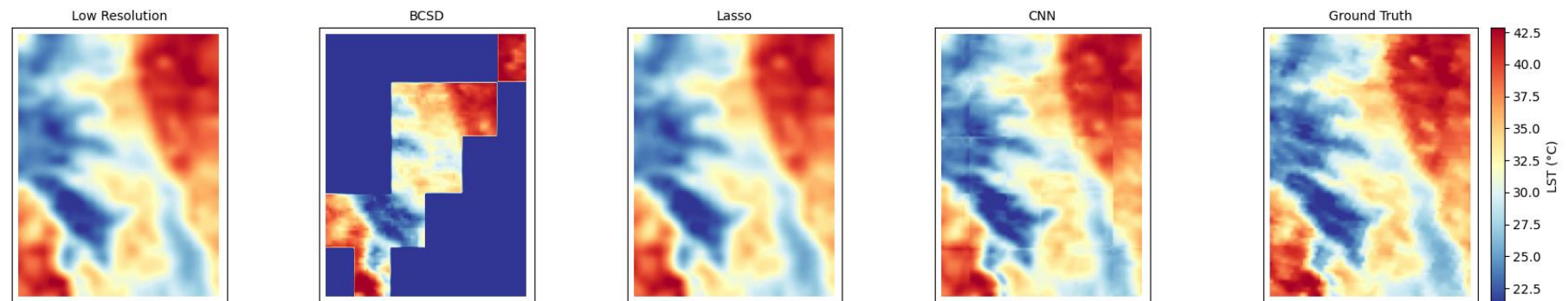
Evaluation Metrics

- **Root Mean Square Error (RMSE)** : Sensitive to large errors or outliers
 - ↓ lower is better
- **Mean Average Error (MAE)**: Treats all errors equally
 - ↓ lower is better
- **R-Squared (R^2)**: Measures how well the data fit the model
 - ↑ higher is better
- **Structural Similarity Index Measure (SSIM)**: Measure perceptual image similarity
 - ↑ higher is better

Baseline Methods

1. LASSO: simple linear model with L1 regularization
2. BCSD: standard climate downscaling method using bias correction + interpolation
3. Base CNN: simple learning-based model that captures local spatial patterns

Method	RMSE	MAE	R^2	SSIM
BCSD	20.335	20.126	-40.3	0.031
LASSO	4.791	3.887	-1.146	0.506
CNN	1.097	0.856	0.944	0.475



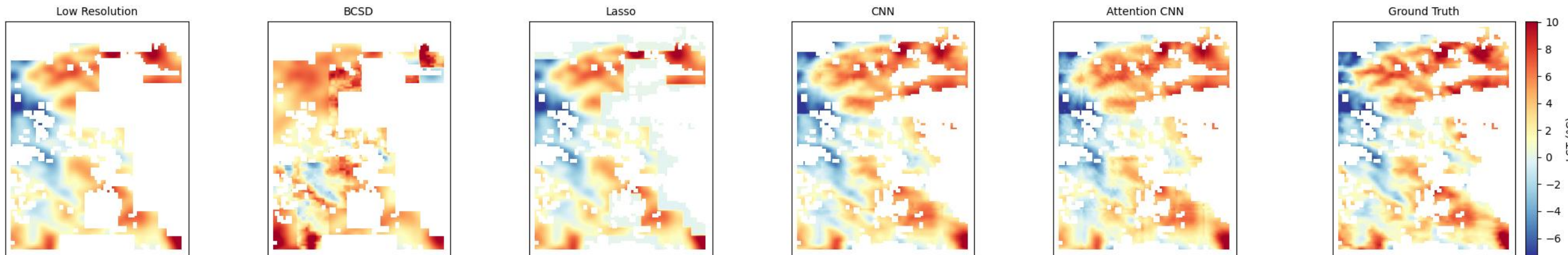
Why These Baselines Matter

- Lasso tests whether simple linear relationships are enough
- BCSD represents a traditional climate downscaling benchmark
 - Relies on past observations at specific training locations
- Baseline CNN shows how much improvement comes from a basic neural model
- Our model is evaluated by how much it improves beyond these baselines

Results (4km to 1km)

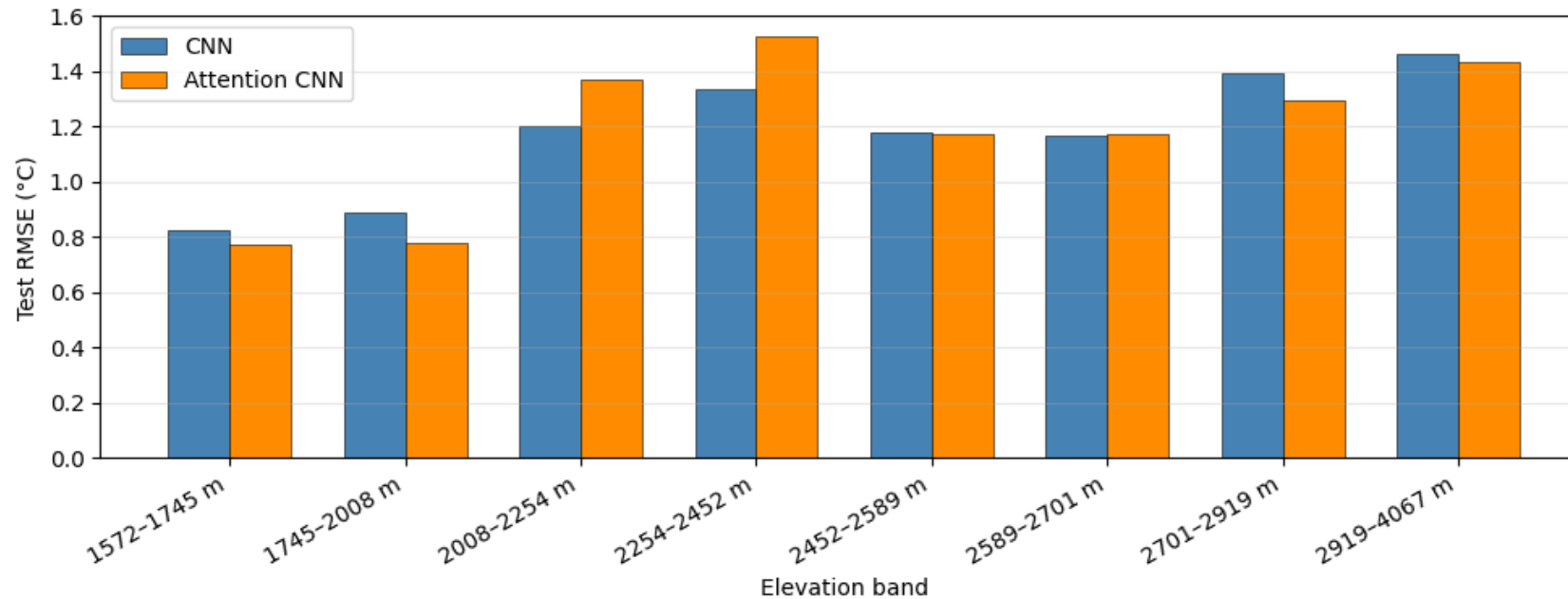
- Our method achieves a higher RMSE and MAE at the expense of a slightly lower R^2 and SSIM

Method	RMSE	MAE	R^2	SSIM
BCSD	20.335	20.126	-40.3	0.031
LASSO	4.791	3.887	-1.146	0.506
CNN	1.097	0.856	0.944	0.475
ATTN-CNN	1.084	0.841	0.941	0.472



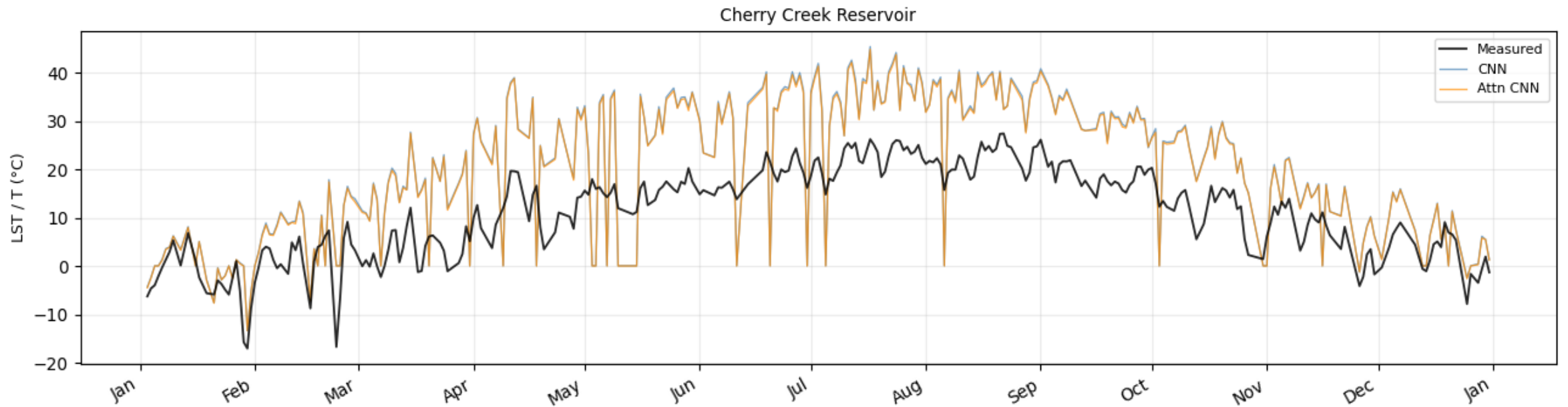
RMSE vs Elevation (4km to 1km)

- Our Attention Augmented CNN outperforms the regular CNN in most elevations bands except between **2008 – 2452 m**



Station Predictions

- The predictions between CNN and attention CNN are indistinguishable from plots
- Both models perform better in the winter and worse during the summer



Station Predictions

- The margin between attention CNN and CNN is pretty indistinguishable
- Will require more time and effort to improve our model

	Station	CNN RMSE (°C)	ATTN-CNN RMSE (°C)
U	Cherry Creek Reservoir	12.52	12.25
U	Chatfield at South Platte Reservoir	10.65	11.03
U	Boulder South West	9.80	9.23
F	Echo Lake	5.67	5.64
F	Jackwhacker Gulch	7.78	7.47
F	Berthoud Summit	7.13	7.24
F	Lake Eldora	4.91	5.05

Conclusions

- Attention augmented CNN based model was able to resolve the 1 km MODIS temperature fields to 250 m.
- Better than the baselines across all elevation ranges except 2008 – 2452 m
- Able to use attention to infer downscaled temperature. Would be better if spatial extent was larger
- The downscaled temperature is accurate for the winter and overestimates the summer peaks
- The model performs better for forested areas as compared to the urban areas.

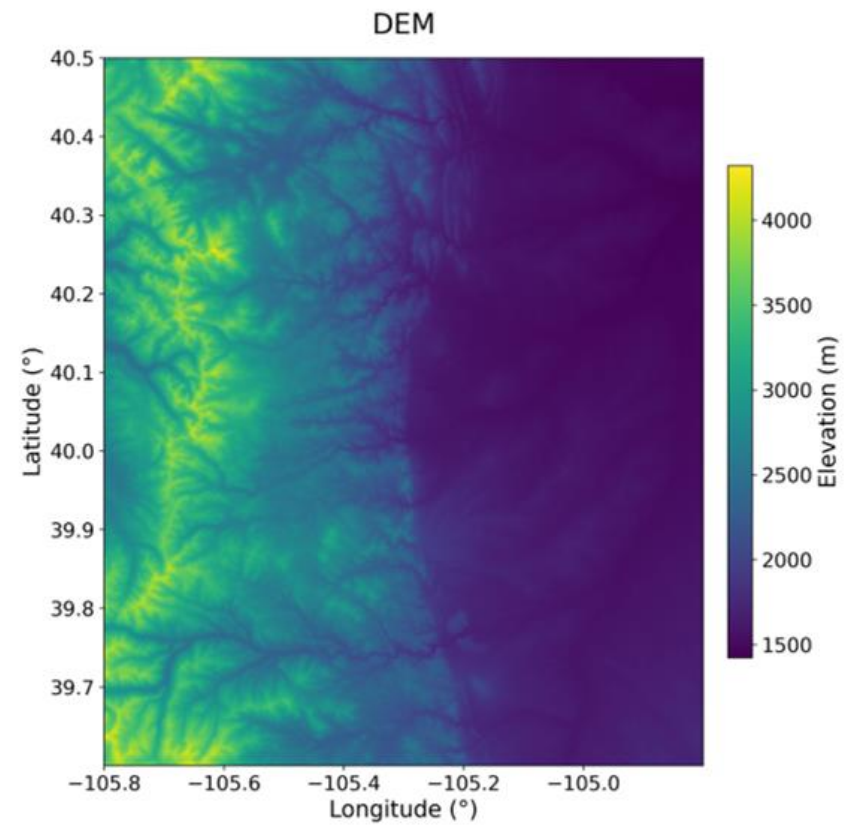
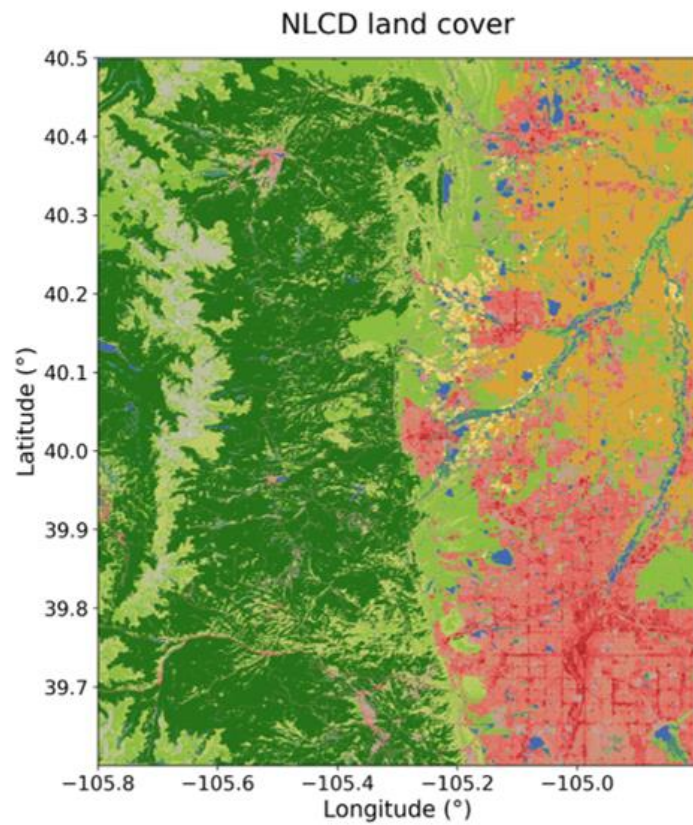
Key Challenges

1. **Sentinel-2** cloud cover issues (forced us to use MODIS NDVI)
2. **Synthetic Domain Gap:** Because we do not have native 250 m thermal satellite data to act as the "ground truth" during training, we have to artificially degrade 1 km MODIS data to 4 km, and train the model to reconstruct the 1 km field.
3. **Spatiotemporal Non-stationarity:** Normally, temperature drops as elevation increases (a standard lapse rate). However, in the Colorado mountains during winter, cold air pools in the valleys, creating intense temperature inversions where the peaks are warmer than the valleys. The network has to learn that the relationship between the DEM and Temperature is not static; it flips based on the season.
4. **Snow Cover Dynamics:** A forest in July has a completely different thermal signature than that exact same forest covered in 3 feet of snow in February. The model must learn to condition its interpretation of the static NLCD land cover map based on the dynamic indices.
5. **Terrain-Induced Shadowing & Topographic Effects:** A steep slope facing direct sunlight will be drastically warmer than a steep slope facing the other direction, which may be entirely in shadow. So, Slope and Aspect, should be included with DEM to reconstruct these microclimates.



Thank you!

Study region



250m Downscaling

- We plot two downscaling methods:
 - Left: Bicubic interpolation from low resolution (4km) to high resolution (250m)
 - Right: Attention CNN prediction where we predict from 4km to 1km and then 1km to 250m.
- Our Attention CNN has more detail although there are notable artifacts near box borders

